

Use Cases Analysis

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Use Case Analysis

Key Ideas

- Use cases are a text-based method of describing and documenting complex processes
- Use cases add detail to the requirements outlined in the requirement definition
- Systems analysts work with users to develop use cases
- Systems analysts develop process and data models later based on the use cases

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Role of Use Cases

- A **use case** is a set of activities that produce some output result
- Describes how the system reacts to an **event** that **triggers** the system
- **Event-driven modeling** – everything in the system is a response to some triggering event

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Role of Use Cases

- All possible responses to the event are documented
- Use cases are helpful when the situation is complicated
- Requirement-gathering techniques: interviews, and observation.

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Elements of a Use Case

- Basic information
 - Name, number and brief description
 - Trigger – event that causes the use case to being
 - External trigger – some from outside the system
 - Temporal triggers – time-based occurrences

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Elements of a Use Case (cont'd)

- Preconditions: Sometimes certain conditions must exist before the use case is valid.
- Interested Stakeholders: The people, organizations, and/or representatives of computer systems who have knowledge necessary to specify this use case or who have an interest in this use case.
- Active Stakeholders: The people, organizations, and/or computer systems that are doing the work of this use case.

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Elements of a Use Case (cont'd)

- Major inputs and outputs
 - Sources and destinations
 - Goal is to be all inclusive
- Details
 - Steps performed and the data inputs and outputs
 - Scenarios

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Elements of a Use Case (cont'd)

- Normal Case Steps: The steps that this use case goes through to complete the normal course of its work.
- Alternatives: Alternatives are acceptable variations on the normal case of processing.
- Exceptions: These are unwanted but necessary variations.
- Outcome: The desired situation at the end of this use case. Think of it as the stake holder's objective at the time when he triggers the use case.

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Scenarios

- The scenario tells the story of a business use case which explain a series of steps for a section of the considered work.
- Business analysts use scenario to describe a business use case to the interested stakeholders.
- Use between three and ten steps to write a scenario.

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Scenarios (cont'd)

A business use case for checking a passenger in for and international flight



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Scenarios (cont'd)

1. Get the passenger's ticket or record locator.
2. Is this the right passenger, flight, and destination?
3. Check the passport is valid and belongs to the passenger.
4. Record the frequent-flyer number.
5. Find a seat.
6. Ask security questions.
7. Check the baggage onto the flight.
8. Print and hand over the boarding pass and bag tags.
9. "Have a nice flight."

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Scenarios (cont'd)

- Stakeholders are invited to participate and revise the scenario until it represents a consensus view of what the work should be
- Formalize the scenario draft
- The first part of the formalized scenario identifies it and gives it meaningful name.
 - Business Use Case Name: Check passenger onto flight
- Next, add the start-up mechanism, or trigger, for the business use case. It usually consists of some data or request (often both) arriving from outside the work area.

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Scenarios (cont'd)

- Trigger: Passenger's ticket, record locator, or identity and flight
- Add any preconditions that must exist when the business use case is triggered.
 - Precondition: The passenger must have a reservation.
- Add the interested stakeholders to the scenario. These people have an interest in the outcome of the business use case.
 - Interested Stakeholders: Check-in agent, marketing, baggage handling, reservations, flight manifest system, work flow, security, destination country's immigration, passenger

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Scenarios (cont'd)

- Among the interested stakeholders, identify the active stakeholders. Active stakeholders are the people or systems that do the work of the business use case or trigger the business use case.
 - Active stakeholder: Passenger (trigger), check-in agent

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Normal Case Scenarios

- Assume for the normal case that everything works perfectly
- Go back to the sketch of scenarios, change the language to be technologically neutral.
 1. Get the passenger's ticket, record locator, or identity and flight number
 1. Locate the passenger's reservation.
 2. Is this the right passenger's reservation
 2. Ensure the passenger is correctly identified and connected to the right reservation

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Normal Case Scenarios (cont'd)

3. Check the passport is valid and belongs to the passenger
3. Check the passport is valid and belongs to the passenger
 - 3.1 The passport must be current
 - 3.2 The passport must not expire before the end of the complete trip.
 - 3.3 The passport must be valid for travel to the destination country
 - 3.4 Visas (where needed) must be current
 - 3.5 There must be no "refused entry" stamps from the destination country

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Normal Case Scenarios (cont'd)

3. Check the passport is valid and belongs to the passenger. See procedure guidelines EU-175

.... and so on.....

- Add outcome to the scenario. Outcome is the desired situation at the successful conclusion of the business use case. Think of it as the stakeholders' objectives at the time when he triggers the use case.

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Business Event: Passenger decides to check in.

Business Use Case Name and Number: Check passenger onto flight.

Trigger: Passenger's ticket, record locator, or identity and flight.

Preconditions: The passenger must have a reservation.

Interested Stakeholders: Check-in agent, marketing, baggage handling, reservations, flight manifest system, work flow, security, destination country's immigration.

Active Stakeholders: Passenger (trigger), check-in agent.

1. Locate the passenger's reservation.
2. Ensure the correct passenger is connected to the reservation.
3. Check the passport is valid and belongs to the passenger. See procedure guidelines EU175.
4. Attach passenger's frequent-flyer number to the reservation.
5. Allocate a seat.
6. Get correct responses to the security questions.
7. Check the baggage onto the flight.
8. Create and convey to the passenger the boarding pass and bag tags.
9. Wish the passenger a pleasant flight.

Outcome: The passenger is recorded as checked onto the flight, the bags are assigned to the flight, a seat is allocated, and the passenger is in possession of a boarding pass and bag claim stubs.

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Alternative Cases

- Alternative Cases arise when there is an intentional choice of user actions
- These cases usually exist to make the work of the business use case more attractive and convenient to the participants.
- Find alternatives by examining each step of the normal case. Look for instances where the step may be carried out differently or the actor can be given a choice.

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Alternative Cases (cont'd)

4. Attach passenger's frequent-flyer number to the reservation.
 - A4.1 Allow the FF number to be changed to that of a partner airline
 - A4.2 Allow the FF number to be changed to that of a family member, or the mileage of the flight to be donated to a charity of the passenger's choice.

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Exception Cases

- Exceptions are unwanted but acceptable deviations from the normal case.
- For example, an end user could enter the wrong data, or the online customer might forget his password.
- The goal of the exception scenario is to show how the work safely handles the exception.
- Add the exception step to the normal scenario

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Exception Cases (cont'd)

5. Find a seat.

E5.1 The passenger's choice of seat is not available.

E5.1 Record a request for a seat change by the gate agent.

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Exception Cases (cont'd)

- find the exceptions by examining each step of the normal case and asking questions such as these:
 - What happens if this step cannot be completed, or does not complete, or comes up with the wrong or unacceptable result?
 - What can go wrong at this step?
 - What could happen to prevent the work from reaching this step?
 - Could any external entity disrupt or prevent this step, or this business use case?
 - Could the end user fail to understand what is required of him, or misunderstand the information presented by the product?
 - Could the end user take the wrong action intentionally or unintentionally or fail to respond?

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Business Use Cases

- A business use case is a preplanned response to a business event
- The business use case is a collection of processes, data that is retrieved and/or stored, output generated, message sent, or some combination of these.
- The business use case is a unit of functionality
- The work can be partitioned by identifying business use cases.
- Use cases are applied for detailed requirements investigation

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Product Use Cases

- A product use case is a part of the business use case and found by examining what the work is and how the work responds to outside demands.
- The product use cases are derived from the business events that are the driving force of the work.
- Product use cases are the processes of business use case handled by the proposed product.
- Whatever is immediately outside the scope of the product become the actor.
- The actor manipulates the functionality of the product use case within the product

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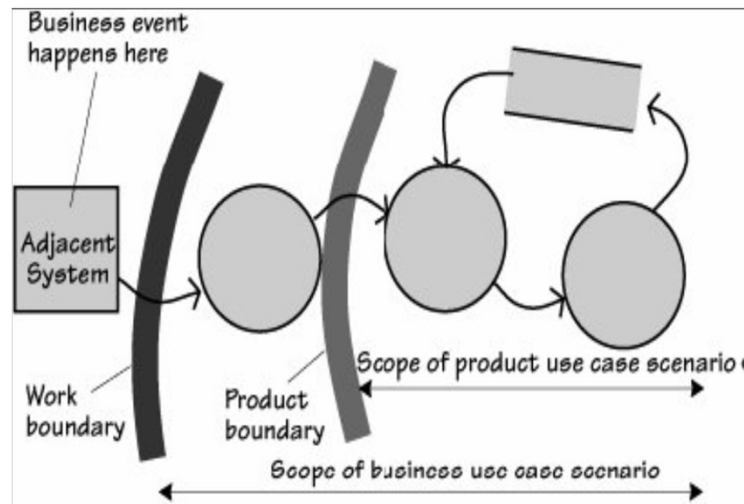
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Product Use Case Scenarios

- When understand enough about the work and its constraints, decide which parts of the work will be carried out by the product.
- The product use case sets out the functionality of the product.
- The difference between business use case and product use case is that the business use case contains all of the business rules that respond to a business event, whereas the corresponding product use case contains only those business rules to be implemented in the product.

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Product Use Case Scenarios (cont'd)

- Identify business event
- Identify business use case
- Write business use case scenario
- Determine how much of the business use case will be implemented as the product.

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Business Event: Passenger decides to check in.

Business Use Case Name and Number: Check passenger onto flight.

Trigger: Passenger's ticket, record locator, or identity and flight.

Preconditions: The passenger must have a reservation.

Interested Stakeholders: Check-in agent, marketing, baggage handling, reservations, flight manifest system, work flow, security, destination country's immigration.

Active Stakeholders: Passenger (trigger), check-in agent.

1. Locate the passenger's reservation.
2. Ensure the correct passenger is connected to the reservation.
3. Check the passport is valid and belongs to the passenger. See procedure guidelines EU175.
4. Attach passenger's frequent-flyer number to the reservation.
5. Allocate a seat.
6. Get correct responses to the security questions.
7. Check the baggage onto the flight.
8. Create and convey to the passenger the boarding pass and bag tags.
9. Wish the passenger a pleasant flight.

Outcome: The passenger is recorded as checked onto the flight, the bags are assigned to the flight, a seat is allocated, and the passenger is in possession of a boarding pass and bag claim stubs.

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Product Use Case Name: Check passenger onto flight.

Trigger: Passenger's name + passenger's passport number.

Preconditions: The passenger must have a reservation.

Interested Stakeholders: Check-in agent, marketing, baggage handling, reservations, flight manifest system, work flow, security, destination country's immigration.

Actor: Check-in agent.

1. Locate the passenger's reservation.
4. Attach the frequent-flyer number to the reservation.
5. Allocate a seat.
6. Check the baggage onto the flight.
7. Print the boarding pass and bag tags.

Outcome: The passenger is recorded as checked onto the flight, the bags are assigned to the flight, a seat is allocated, and the passenger's boarding pass and bag claim stubs are printed.

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Product Use Case Scenarios (cont'd)

- Instead of a list of active stakeholders, the check-in agent is identified as an actor.
- An actor is a person or system that has a direct interface with the product.

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Use Case Diagram

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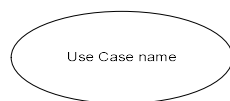
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Use Case Diagram

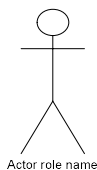
- Summarizes all use cases (for the part of the system being modeled) together in one picture
- Typically drawn early in the SDLC
- Shows the associations between actors and use cases

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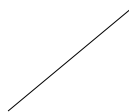
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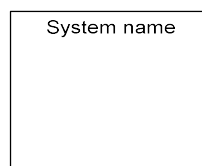
Use case: represent a major piece of system functionality



Actor: a person or system that derives benefit from and is external to the system



Communication line: link an actor with the use case(s) with which it interact



System boundary: includes the name of the system inside or on top
Represents the scope the system

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Use case

The important thing to remember is that a use case from the user's perspective is a complete use of the system

Communication Lines

- A communication line connects an actor and a use case to show the actor participating in the use case
- The purpose of a communication line is to show that an actor is simply involved in a use case, not to imply an information exchange in any particular direction or that the actor starts the use case.

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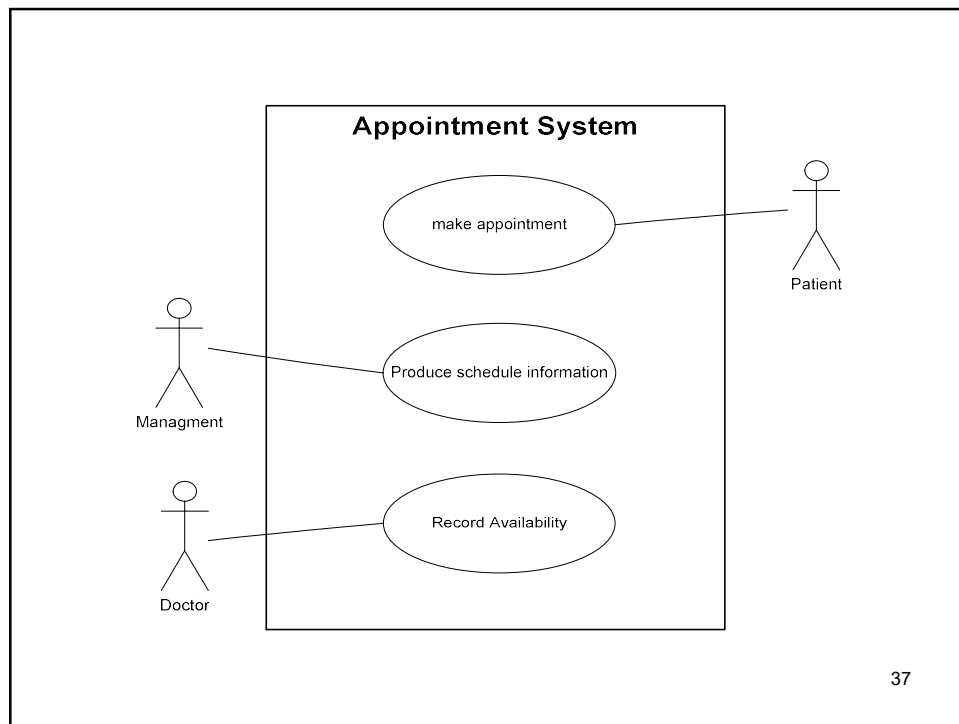
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System boundary

To show your system's boundary on a use case diagram, draw a box around all of the use cases but keep the actors outside of the box.

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Use Case Relationship

- There is some similarity between steps in different use cases.
- Get rid of the repetition between use case descriptions.
- Show reusable and optional use case behavior between use cases.

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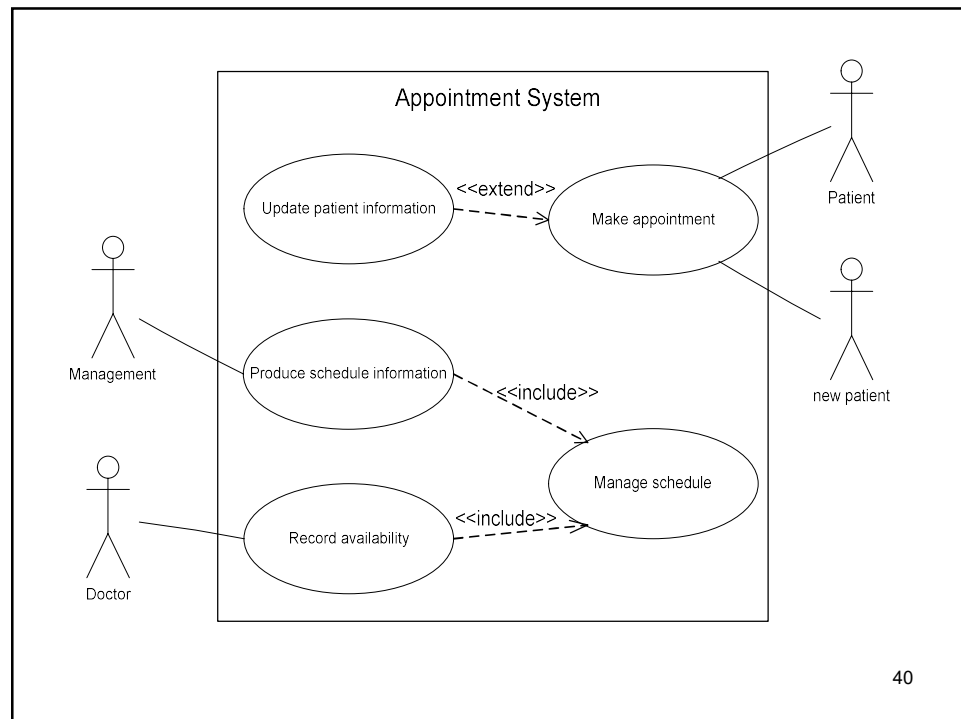
Use Case Relationship (cont'd)

The <<include>> Relationship

The <<include>> relationship declares that the use case at the tail of the dotted arrow completely reuses all of the steps from the use case being included.

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Use Case Relationship (cont'd)

The <<extend>> Relationship

The <<extend>> relationship shows that a use case might completely reuse another use case's behavior, similar to the <<include>> relationship, but that this reuse was optional and dependent either on a runtime or system implementation decision.

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Steps in Creating the Use Case Diagram

1. Identify use cases
2. Draw the system boundary
3. Place use cases on the diagram
4. Identify the actors
5. Add communication lines

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Q&A

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